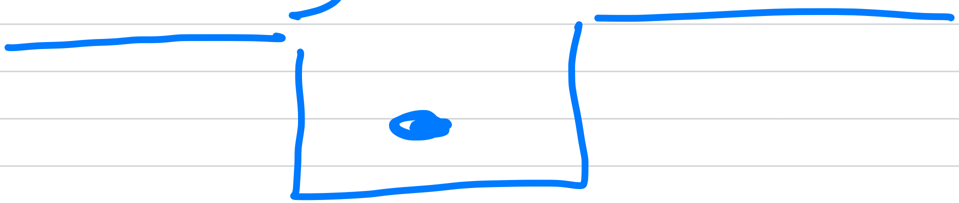
FYS4480/9480 AUG 20

Hamiltonian $\hat{H} = \hat{H}_0 + \hat{H}_I$

non-interacting part (approx and can be solved analytically)

kinetic energy + $V_{ext}(x)$



two-body or higher-body interaction

$$\hat{H}|\psi_i\rangle = E_i|\psi_i\rangle$$

$$\hat{H}_0|\Phi_j\rangle = \varepsilon_i|\Phi_j\rangle$$

$$\hat{H}|\Phi_j\rangle \neq E_j|\Phi_j\rangle$$

Φ_j is an ONB +

$$\langle\Phi_j|\Phi_i\rangle = \delta_{ij}$$

$$|\psi_i\rangle = \sum_j c_{ij}|\Phi_j\rangle$$

$$c_{ij} = \langle \psi_i | \Phi_j \rangle$$

The sum in

$$|\psi_i\rangle = \sum_{j=0}^{\infty} c_{ij} |\Phi_j\rangle$$

c_{ij} are the matrix elements of a unitary matrix C

$$CC^{+} = C^{+}C = \mathbb{1}$$

$$C^{+} = C^{-1}$$

$$\sum_{j=0}^{\infty} c_{ij} |\Phi_j\rangle \simeq \sum_{j=0}^{d} c_{ij} |\Phi_j\rangle$$

d = dimension of effective Hilbert space

$$C \in \mathbb{C}^{d \times d}$$

$$|\Psi_i\rangle \Rightarrow |\Psi\rangle = C |\Phi\rangle$$

Two-level system

$\uparrow$ ε_1 $\quad |\phi_1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ computational basis

$\downarrow$ ε_0 $\quad |\phi_0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$\mathcal{H}_0 |\Phi_0\rangle = \varepsilon_0 |\Phi_0\rangle \qquad \langle \phi_0 | \phi_1 \rangle$$

$$\mathcal{H}_0 |\phi_1\rangle = \varepsilon_1 |\Phi_1\rangle \qquad = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$= 0$$

$$\mathcal{H}_I \Rightarrow \mathcal{H} |\psi_0\rangle \sim \mathcal{H} |\psi_i\rangle$$

$$|\psi_0\rangle = \underbrace{\alpha}_{c_{00}} |\Phi_0\rangle + \underbrace{\beta}_{c_{01}} |\Phi_1\rangle$$

$$\det(\mathcal{H} - \lambda I) = 0$$

$$|\psi_1\rangle = \underbrace{\gamma}_{c_{10}} |\Phi_0\rangle + \underbrace{\delta}_{c_{11}} |\Phi_1\rangle$$

$$\underline{1} = \sum_{i=0}^{1} |\Phi_i\rangle\langle\Phi_i| = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$|\Phi_0\rangle\langle\Phi_0| \qquad \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix}$$

$$\vec{\Phi}_0 \vec{\Phi}_0^T$$

$$\hat{P} = |\Phi_0\rangle\langle\Phi_0| \wedge \hat{Q} = |\Phi_1\rangle\langle\Phi_1|$$

$$\hat{P}^2 = \hat{P} \wedge \hat{Q}^2 = \hat{Q}$$

$$[\hat{P}, \hat{Q}] = \hat{P}\hat{Q} - \hat{Q}\hat{P} = 0$$

$$\hat{P} \cdot \hat{Q} = 0$$

$\hat{P}$ and $\hat{Q}$ are projection operators

$$\hat{P}\,|\Psi_0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\left(\alpha\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \beta\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right)$$

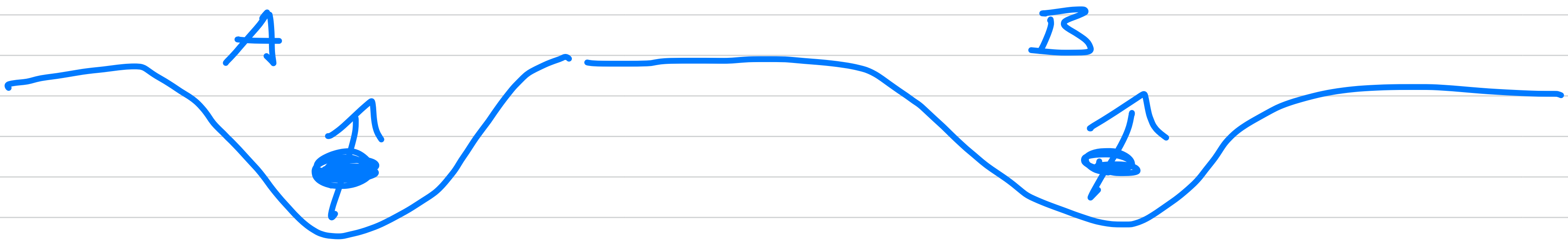
$$= \alpha\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \alpha\,|\Phi_0\rangle$$

$$\hat{Q}\,|\Psi_0\rangle = \beta\,|\Phi_1\rangle$$

Two-"particle" computational basis

$$\begin{bmatrix}0\\1\end{bmatrix}_A \quad \varepsilon_1^A \qquad\qquad \begin{bmatrix}0\\1\end{bmatrix}_B \quad \varepsilon_1^B$$

$$\begin{bmatrix}1\\0\end{bmatrix}_A \quad \varepsilon_0^A \qquad\qquad \begin{bmatrix}1\\0\end{bmatrix}_B \quad \varepsilon_0^B$$



$$|\Phi_0\rangle_{AB} = |\Phi_0\rangle_A \otimes |\Phi_0\rangle_B$$

$$= \begin{bmatrix}1\\0\end{bmatrix} \otimes \begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}1\\0\\0\\0\end{bmatrix}$$

$$H_0 |\Phi_0\rangle_{AB} = \varepsilon_0^{AB} |\Phi_0\rangle_{AB}$$

$$\begin{bmatrix}1\\0\\0\\0\end{bmatrix} = |00\rangle \quad \left(= |0\rangle \right)$$

$$\left(|x_0 x_1\rangle \to 2\underset{0}{x_0} + \underset{0}{x_1} = 0 \right.$$

$$|\Phi_1\rangle_{AB} = |\Phi_0\rangle_A \otimes |\Phi_1\rangle_B$$

$$= \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$= |01\rangle \quad (|1\rangle)$$

$$2x_0 + x_1 = 1$$
$$\quad 0 \qquad 1$$

$$|\Phi_2\rangle_{AB} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = |10\rangle \quad (|2\rangle)$$

$$|11\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}_A \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix}_B$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \quad (= |3\rangle$$

$2x_0 + 1$

$1 \quad 1 = 3$

$$\langle 00 | 01 \rangle = 0$$

ONB

$$\mathcal{H} |\psi_0\rangle = \alpha |00\rangle + \beta |01\rangle$$
$$+ \gamma |10\rangle + \delta |11\rangle$$